

(a) Shared excerpts**Inverse queries exchange the correct answer**

$q_+ : \text{dog followed by rooster} \rightarrow W_+$

$q_- : \text{rooster followed by dog} \rightarrow W_-$

(b) RelTwin: two-stage fine-tuning

TRAINING

1 Query-conditioned competition

Score complete timestamp answers

	$y(W_+)$	$y(W_-)$
q_+	$s_{++} \uparrow$	$s_{+-} \downarrow$
q_-	$s_{-+} \downarrow$	$s_{--} \uparrow$

Candidate CE + label-exchange JS

2 Set-level preference learning

Compare coverage, count, and boundaries

Quality scores $\rightarrow$ soft preference targets**Inference**Audio + query $\rightarrow$ Adapted backbone $\rightarrow$ Timestamps**(c) Query-specific selection**

Inverse-query grounding example

Paired SFT**+ Candidate**